Week 2 Assignment Solutions

5. Stop-Word-Filtered Word Frequency Report

Lowercase the paragraph, remove periods and commas, ignore the specified stop words, and display word counts from highest to lowest.

Java Solution

```
import java.util.*;

public class Main {
    static void printFilteredWordFrequency(String feedback) {
        Set<String> stopWords = new HashSet<>(Arrays.asList(
                "the", "was", "and", "a", "is", "of", "in"));
        Map<String, Integer> frequency = new HashMap<>();

        String cleaned = feedback.toLowerCase()
                .replace(".", "")
                .replace(",", "");
        String[] words = cleaned.trim().split("\\s+");

        for (String word : words) {
            if (!word.isEmpty() && !stopWords.contains(word)) {
                frequency.put(word, frequency.getOrDefault(word, 0) + 1);
            }
        }

        List<Map.Entry<String, Integer>> entries =
                new ArrayList<>(frequency.entrySet());
        entries.sort((a, b) -> b.getValue().compareTo(a.getValue()));

        for (Map.Entry<String, Integer> entry : entries) {
            System.out.println(entry.getKey() + ": " + entry.getValue());
        }
    }

    public static void main(String[] args) {
        printFilteredWordFrequency(
                "The mentor was great, the session was great and clear.");
    }
}
```

Sample Output

```
great: 2
mentor: 1
session: 1
clear: 1
```

Notes

Only periods and commas are removed, matching the prompt. Words tied at the same frequency may appear in any order.